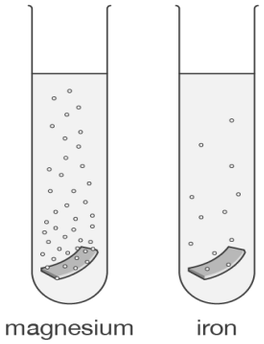
Knowledge Organiser
Year 8: Chemical Reactions - Reactions of Metals

Reactions of metals and acids:
- When metals react with an acid, a **metal salt** and **hydrogen gas** are formed
- A metal salt is a metal/non-metal **compound** formed from an acid
- The hydrogen gas will be observed **bubbling** out of the reaction mixture



The hydrogen rises up out of the reaction mixture because it has a **lower density** than the acid.

General word equation for the reaction of metals and acids:

ACID + METAL ⟶ SALT + HYDROGEN

Different types of salts?
- Depending on the acid used, a different type of **salt** can be formed
- Hydrochloric acid (HCl) forms **chloride salts**
- Sulfuric acid (H_2SO_4) forms **sulfate salts**
- Nitric acid (HNO_3) forms **nitrate salts**

Reactants	Products
Zinc + Hydrochloric acid	Zinc chloride + Hydrogen
Tin + Nitric acid	Tin nitrate + Hydrogen
Iron + Sulfuric acid	Iron sulfate + Hydrogen

Reactions of metal oxides:
- Metal oxides are compounds containing a metal bonded to oxygen
- Metal oxides are bases, so undergo neutralisation reactions with acids
- When metal oxides react with an acid, a metal salt and water are formed

General word equation for the neutralisation reaction of metal oxides (a base) and acids

ACID + BASE ⟶ SALT + WATER

Reactants	Products
Titanium oxide + Nitric acid	Titanium nitrate + Water
Tin oxide + Hydrochloric acid	Tin chloride + Water
Lead oxide + Sulfuric acid	Iron sulfate + Water

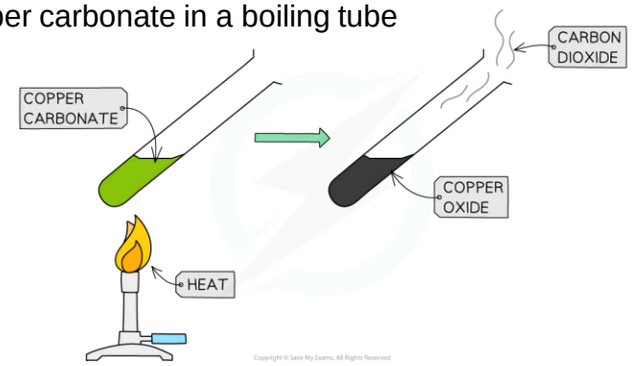
The reactivity series:
- Metals make up 80% of all known elements, their chemical properties are varied
- Some metals are **so reactive** that will ignite when placed in water e.g. potassium
- Some metals are **so unreactive** that they will not react with concentrated acid e.g. gold
- We can place metals in order of their reactivity in a **reactivity series**

The reactivity series of metals, with a mnemonic (memory device) to help you remember the order



Thermal decomposition of metal carbonates:
- Metal carbonates are compounds which contain a metal, bonded to oxygen and carbon e.g. $MgCO_3$ (magnesium carbonate)
- These compounds will undergo thermal decomposition
- Thermal (heat) decomposition (breaking down) just means the compound will break apart under heating
- In this reaction a metal oxide and carbon dioxide are formed

A diagram showing the thermal decomposition of copper carbonate in a boiling tube

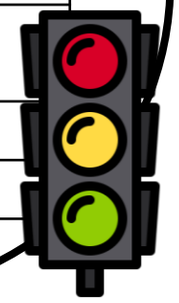


In these reactions there is only one reactant as the heat is breaking it down.

Reactants	Products
Magnesium carbonate	Magnesium oxide + Carbon dioxide
Sodium carbonate	Sodium oxide + Carbon dioxide
Silver carbonate	Silver oxide + Carbon dioxide

RAG Yourself!

Learning Outcomes;	RAG
State the general word equation for the reaction of acids and metals	
Describe the thermal decomposition of metal carbonates	
State the general word equation for the reaction of acids and metal oxides	
State the order of metals in the reactivity series	
Explain why gold and silver are used for jewellery	



Knowledge Organiser
Year 8: Chemical Reactions - Reactions of Metals

Reactions of metals and acids:

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General word equation for the reaction of metals and acids:

Different types of salts?

Reactants	Products
Zinc + Hydrochloric acid	
Tin + Nitric acid	
Iron + Sulfuric acid	

Reactions of metal oxides:

General word equation for the neutralisation reaction of metal oxides (a base) and acids

Reactants	Products
Titanium oxide + Nitric acid	
Tin oxide + Hydrochloric acid	
Lead oxide + Sulfuric acid	

The reactivity series:

The reactivity series of metals, with a mnemonic (memory device) to help you remember the order

Thermal decomposition of metal carbonates:

A diagram showing the thermal decomposition of copper carbonate in a boiling tube

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Sodium carbonate	
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